

# 2.3 Group 17

## Question Paper

|            |                        |
|------------|------------------------|
| Course     | CIEA Level Chemistry   |
| Section    | 2. Inorganic Chemistry |
| Topic      | 2.3 Group 17           |
| Difficulty | Hard                   |

**Time allowed:** 40

**Score:** /29

**Percentage:** /100

### Question 1a

Silver nitrate solution,  $\text{AgNO}_3(\text{aq})$ , is added to separate solutions of  $\text{NaI}$  and  $\text{NaCl}$  and precipitates form. An excess of aqueous ammonia is then added to both precipitates.

i)

Complete Table 1.1 to give the colour and name of the precipitate formed in each reaction and the effect of the addition of an excess of aqueous ammonia to each of the precipitates formed.

Table 1.1

|                                                                       | $\text{NaI}(\text{aq}) + \text{AgNO}_3(\text{aq})$ | $\text{NaCl}(\text{aq}) + \text{AgNO}_3(\text{aq})$ |
|-----------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| Colour of precipitate                                                 |                                                    |                                                     |
| Name of precipitate                                                   |                                                    |                                                     |
| Effect of addition of an excess of aqueous ammonia to the precipitate |                                                    |                                                     |

[3]

ii)

Write an ionic equation, including state symbols, to show the reaction occurring when  $\text{AgNO}_3(\text{aq})$  is added to  $\text{NaI}(\text{aq})$ .

[1]

[4 marks]

### Question 1b

Solid  $\text{NaI}$  reacts with concentrated sulfuric acid to form purple fumes of  $\text{I}_2(\text{g})$  and hydrogen sulfide gas,  $\text{H}_2\text{S}(\text{g})$ . However, when solid  $\text{NaCl}$  reacts with concentrated sulfuric acid the only gas produced is  $\text{HCl}(\text{g})$ .

Explain the difference in the reactions of concentrated sulfuric acid with  $\text{NaI}$  and with  $\text{NaCl}$ . Your answer should refer to the role of the sulfuric acid in each reaction.

[3 marks]

### Question 1c

Chlorine is commonly used in water purification. When chlorine is added to water it reacts to produce a mixture of acids, one of which is chloric(I) acid,  $\text{HClO}$ , a powerful oxidising agent.

i)

Explain the meaning of the term oxidising agent, in terms of electron transfer.

[1]

ii)

Suggest an equation for this reaction of chlorine with water.

[1]

iii)

Write an equation for the reaction of chlorine with hot aqueous sodium hydroxide.

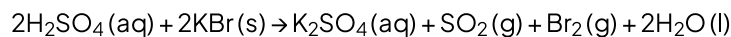
Use oxidation numbers to explain why this is a redox reaction.

[2]

**[4 marks]**

### Question 2a

An acid base reaction occurs for the reaction between solid potassium bromide and concentrated sulfuric acid. The equation for this is:



In this reaction redox products are also formed. State the **two** redox products and possible observation for each one.

Redox product 1 .....

Observation .....

Redox product 2 .....

Observation .....

[2 marks]

### Question 2b

Explain why the reaction between solid potassium iodide and concentrated sulfuric acid produces hydrogen sulfide whereas the reaction between solid potassium bromide, and concentrated sulfuric acid does not.

[4 marks]

### Question 2c

Hydrogen halides,  $\text{H-X}$ , are formed from the reaction of hydrogen and a halogen,  $\text{X}_2$ . Explain why in solution hydrogen fluoride is classed as a weak acid, whereas hydrogen chloride is classed as a strong acid.

[2 marks]

### Question 3a

The halogens can react with iron wool to produce various iron compounds including iron(III) chloride, iron(III) bromide and iron(II) iodide.

The  $\text{Fe}^{2+}$  ion can also react with chlorine,  $\text{Cl}_2$  and bromine,  $\text{Br}_2$ .

i)

State the half-equation for the conversion of  $\text{Br}_2$  to  $\text{Br}^-$

[1]

ii)

State the half-equation for the conversion of  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$

[1]

iii)

Hence, give the full equation for the reaction of  $\text{Br}_2$  and  $\text{Fe}^{2+}$

[1]

iv)

Explain why  $\text{I}_2$  cannot convert  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$

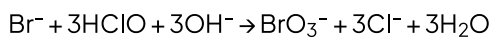
[1]

[4 marks]

### Question 3b

Bromate ions are toxic and fish eggs exposed to bromate can develop disorders in the brain and spine. This can also poison humans, where symptoms include acute gastrointestinal irritation and depression of the central nervous system.

Another reaction in which bromate ions are formed is by the reaction with disinfection contaminants, such as hypochlorite,  $\text{HClO}$ , in sea water shown in the equation.



State the oxidation number for Cl in  $\text{HClO}$  and  $\text{Cl}^-$ .

[2 marks]

### Question 3c

Identify the oxidising agent in the reaction given in part (c). Explain your answer.

[2 marks]

### Question 4

Table 3.1 shows the boiling points of some halogenoalkanes.

Table 3.1

| Compound                                     | Boiling Point / °C |
|----------------------------------------------|--------------------|
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{F}$  | -2.3               |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ | 46.6               |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ | 71.0               |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{I}$  | 102.5              |

Explain the trend in boiling points of the halogenoalkanes in Table 3.1.

[2 marks]



## Model Answers: Hard

1a

a)

i) Completed Table 1.1:

|                                                                              | <b><math>\text{NaI (aq)} + \text{AgNO}_3 \text{ (aq)}</math></b> | <b><math>\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)}</math></b> |
|------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Colour of precipitate</b>                                                 | yellow                                                           | white; [1 mark]                                                   |
| <b>Name of precipitate</b>                                                   | silver iodide                                                    | silver chloride; [1 mark]                                         |
| <b>Effect of addition of an excess of aqueous ammonia to the precipitate</b> | no (visible) change                                              | dissolves / soluble; [1 mark]                                     |

ii) The ionic equation for the reaction occurring when  $\text{AgNO}_3 \text{ (aq)}$  is added to  $\text{NaI (aq)}$  is:

- $\text{Ag}^+ \text{ (aq)} + \text{I}^- \text{ (aq)} \rightarrow \text{AgI (s)}$ ; [1 mark]

**[Total: 4 marks]**

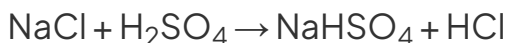
- You must learn the tests for halides very thoroughly
  - You could be asked to apply your knowledge to an unknown situation, without it specifically telling you that you are testing for halide ions
  - You could be asked to provide a full, detailed method to test for halides
  - You will usually be expected to provide the simplest ionic equations for these reactions – include state symbols, even if not specified, as this is an example of a precipitation reaction and state symbols may be required for the mark

1b

b) The difference in the reactions of concentrated sulfuric acid with  $\text{NaI}$  and with  $\text{NaCl}$  is:

- Sulfuric acid acts as an acid with  $\text{NaCl}$   
**OR**





**OR**



- Sulfuric acid acts as an oxidising agent with NaI / I<sup>-</sup>

**OR**

NaI is a reducing agent; [1 mark]

- I<sup>-</sup> more powerful reducing agent than Cl<sup>-</sup>

**OR**

Sulfuric acid can oxidise I<sup>-</sup> but not Cl<sup>-</sup>

**OR**

Sulfuric acid is a stronger oxidising agent than iodide ions

**OR**

Sulfuric acid is not as strong an oxidising agent than chloride ions; [1 mark]

**[Total: 3 marks]**

- Depending on the halide ion in the sodium halide the reaction of concentrated sulfuric acid forms a number of different products
- The only reaction that occurs with NaCl and concentrated sulfuric acid is an acid-base reaction
- This is because chloride is not a strong enough reducing agent to reduce the sulfur in sulfuric acid
- As you go down the group the reducing ability increases
- Therefore redox products can be formed with NaBr and NaI
- You must be clear on the difference between oxidising ability and reducing ability
  - Oxidising ability is the ability of a species to oxidise a different species (i.e. how easily it can gain electrons)
  - Reducing ability is the ability of a species to reduce a different species (i.e. how easily it can lose electrons)
- The reducing ability of the halides increases moving down the group
  - This is because the size of the ions increases therefore it is easier to donate an electron

1c

c)

i) The meaning of the term oxidising agent, in terms of electron transfer is:

- (Species that) gains electrons / electron acceptor; [1 mark]

ii) An equation for this reaction of chlorine with water is:

- $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HClO}$ ; [1 mark]

iii) An equation for the reaction of chlorine with hot aqueous sodium hydroxide is

- $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$ ; [1 mark]

It is a redox equation because:

- Oxidation number of chlorine changes from 0 to -1 and (+)5

**AND**

Chlorine has been oxidised and reduced; [1 mark]

**[Total: 4 marks]**

- Make sure you know the definitions for oxidising agent as well as reducing agent in terms of electrons
  - An oxidising agent is itself reduced, so will gain electrons
  - A reducing agent is itself oxidised, so will donate electrons
- You should know the equations for the reaction of chlorine with both hot and cold sodium hydroxide, there are subtle differences between the two equations
- Hot
  - $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$
- Cold
  - $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$
- When explaining whether a reaction is a redox reaction, make sure you use oxidation numbers to explain
  - If the oxidation number increases oxidation has occurred
  - If the oxidation number decreases then reduction has occurred
- The oxidation number of chlorine in  $\text{Cl}_2$  is 0
- The oxidation number of chlorine in  $\text{NaCl}$  is -1 as Na is +1
- The oxidation number of chlorine in  $\text{NaClO}_3$  is +5
  - Each oxygen is -2 and there are 3 O atoms = -6
  - The Na atom has an oxidation number of +1
  - $\text{NaClO}_3$  has no charge overall so the oxidation number of Cl must be +5

2a

a) The **two** redox products and possible observations for each one are

- $\text{SO}_2$  / sulfur dioxide

**AND**

Colourless gas evolved / no observation; [1 mark]

- $\text{Br}_2$  / bromine

**AND**

Brown / orange vapour / fumes; [1 mark]

**[Total: 2 marks]**

- **Careful:**

- The question only asks for redox products so only list the products if there has been a change in oxidation state

- The redox products are:

- $\text{Br}_2$  and  $\text{SO}_2$  **only**

- $\text{SO}_2$  is a colourless gas but can be tested by holding filter paper soaked in potassium dichromate solution

- The colour change is orange to green

2b

b) Explain why the reaction between solid potassium iodide, KI, and concentrated sulfuric acid,  $\text{H}_2\text{SO}_4$ , produces  $\text{H}_2\text{S}$  whereas the reaction between solid potassium bromide, KBr, and concentrated sulfuric acid,  $\text{H}_2\text{SO}_4$ , does not:

- Iodide ion /  $\text{I}^-$  stronger reducing agent than bromide ion /  $\text{Br}^-$ ; [1 mark]
- Able to reduce S / sulfur in  $\text{H}_2\text{SO}_4$  from +6 to -2 in  $\text{H}_2\text{S}$ ; [1 mark]
- Iodide ion /  $\text{I}^-$  much bigger than bromide ion /  $\text{Br}^-$ ; [1 mark]
- Readily donates an electron (compared to  $\text{Br}^-$ ) (as further away from the nucleus / more shielding); [1 mark]

**[Total: 4 marks]**

- The reducing power (ability to donate electrons) of the halogens reduces down Group 17
- In this example we are talking about a halide ion,  $\text{X}^-$ 
  - As you descend Group 17 the size of the halide ion increases as there are more electron shells

- This means the outermost electron is further away from the positive nucleus so it is much more easily donated
- The sulfur, S, atom in  $\text{H}_2\text{SO}_4$  can therefore be reduced from +6 to -2 in  $\text{H}_2\text{S}$ 
  - $\text{H}_2\text{SO}_4$ ; **2 H atoms** (+2) and **4 O atoms** (-8) mean the oxidation state of S is +6
  - $\text{H}_2\text{S}$ ; **2 H atoms** (+2) mean the oxidation state of S is -2
- The change from +6 to -2 shows that 8 electrons overall have been gained

2c

c) H-F is classed as a weak acid and H-Cl is classed as a strong acid because:

- The H-F bond is stronger / shorter than the H-Cl bond; [1 mark]
- Easier for the H-Cl bond to break (and dissociate releasing  $\text{H}^+$  ions)

**OR**

H-F bond does not break as easily (so fewer  $\text{H}^+$  ions are released); [1 mark]

**[Total: 2 marks]**

- For an acid to be classed as 'strong' it must fully dissociate and therefore release a high number of  $\text{H}^+$  ions
  - $\text{HCl} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
- Weak acids only partially dissociate so do not release many  $\text{H}^+$  ions
- As fluorine is the smallest halogen the H-F bond is the shortest
  - This makes it stronger than H-Cl
  - This also means that H-Cl is a stronger bond than H-Br
- The stronger the bond, the less likely dissociation will occur and  $\text{H}^+$  ions will be released

3a

a)

i) The half-equation for the conversion of  $\text{Br}_2$  to  $\text{Br}^-$  is:

- $\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$ ; [1 mark]

ii) The half-equation for the conversion of  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$  is:

- $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ ; [1 mark]

iii) Hence, give the full equation for the reaction of  $\text{Br}_2$  and  $\text{Fe}^{2+}$

- $\text{Br}_2 + 2\text{Fe}^{2+} \rightarrow 2\text{Br}^- + 2\text{Fe}^{3+}$ ; [1 mark]

iv)  $\text{I}_2$  cannot convert  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$  because:

- Iodine /  $\text{I}_2$  is not a strong enough oxidising agent; [1 mark]

[Total: 4 marks]

- The oxidising power of the halogens **decreases** as you go down the group
- The halogen molecules find it more difficult to accept electrons:
  - This is because the atoms become larger due to extra shells / shielding
  - Distance between the nucleus and outer shell increases so attraction is weaker
  - Becomes more difficult to gain an electron
- The **reducing power** of the halogens **increases** as you go down the group
- The halogen molecules find it easier to donate electrons

- (Species that) gains electrons / electron acceptor; [1 mark]

ii) An equation for this reaction of chlorine with water is:

- $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HClO}$ ; [1 mark]

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- $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$ ; [1 mark]

It is a redox equation because:

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Chlorine has been oxidised and reduced; [1 mark]

**[Total: 4 marks]**

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  - Each oxygen is -2 and there are 3 O atoms = -6
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2a

a) The **two** redox products and possible observations for each one are

- $\text{SO}_2$  / sulfur dioxide

**AND**

Colourless gas evolved / no observation; [1 mark]

- $\text{Br}_2$  / bromine

**AND**

Brown / orange vapour / fumes; [1 mark]

**[Total: 2 marks]**

- **Careful:**

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- The redox products are:

- $\text{Br}_2$  and  $\text{SO}_2$  **only**

- $\text{SO}_2$  is a colourless gas but can be tested by holding filter paper soaked in potassium dichromate solution

- The colour change is orange to green

2b

b) Explain why the reaction between solid potassium iodide, KI, and concentrated sulfuric acid,  $\text{H}_2\text{SO}_4$ , produces  $\text{H}_2\text{S}$  whereas the reaction between solid potassium bromide, KBr, and concentrated sulfuric acid,  $\text{H}_2\text{SO}_4$ , does not:

- Iodide ion /  $\text{I}^-$  stronger reducing agent than bromide ion /  $\text{Br}^-$ ; [1 mark]
- Able to reduce S / sulfur in  $\text{H}_2\text{SO}_4$  from +6 to -2 in  $\text{H}_2\text{S}$ ; [1 mark]
- Iodide ion /  $\text{I}^-$  much bigger than bromide ion /  $\text{Br}^-$ ; [1 mark]
- Readily donates an electron (compared to  $\text{Br}^-$ ) (as further away from the nucleus / more shielding); [1 mark]

**[Total: 4 marks]**

- The reducing power (ability to donate electrons) of the halogens reduces down Group 17
- In this example we are talking about a halide ion,  $\text{X}^-$ 
  - As you descend Group 17 the size of the halide ion increases as there are more electron shells

- This means the outermost electron is further away from the positive nucleus so it is much more easily donated
- The sulfur, S, atom in  $\text{H}_2\text{SO}_4$  can therefore be reduced from +6 to -2 in  $\text{H}_2\text{S}$ 
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2c

c) H-F is classed as a weak acid and H-Cl is classed as a strong acid because:

- The H-F bond is stronger / shorter than the H-Cl bond; [1 mark]
- Easier for the H-Cl bond to break (and dissociate releasing  $\text{H}^+$  ions)

**OR**

H-F bond does not break as easily (so fewer  $\text{H}^+$  ions are released); [1 mark]

**[Total: 2 marks]**

- For an acid to be classed as 'strong' it must fully dissociate and therefore release a high number of  $\text{H}^+$  ions
  - $\text{HCl} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
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- As fluorine is the smallest halogen the H-F bond is the shortest
  - This makes it stronger than H-Cl
  - This also means that H-Cl is a stronger bond than H-Br
- The stronger the bond, the less likely dissociation will occur and  $\text{H}^+$  ions will be released

3a

a)

i) The half-equation for the conversion of  $\text{Br}_2$  to  $\text{Br}^-$  is:

- $\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$ ; [1 mark]

ii) The half-equation for the conversion of  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$  is:

- $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ ; [1 mark]



iii) Hence, give the full equation for the reaction of  $\text{Br}_2$  and  $\text{Fe}^{2+}$



iv)  $\text{I}_2$  cannot convert  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$  because:

- Iodine /  $\text{I}_2$  is not a strong enough oxidising agent; [1 mark]

[Total: 4 marks]

- The oxidising power of the halogens **decreases** as you go down the group
- The halogen molecules find it more difficult to accept electrons:
  - This is because the atoms become larger due to extra shells / shielding
  - Distance between the nucleus and outer shell increases so attraction is weaker
  - Becomes more difficult to gain an electron
- The **reducing power** of the halogens **increases** as you go down the group
- The halogen molecules find it easier to donate electrons
  - This again is due to the size of the atom
  - If the outer shell is further away from the nucleus it becomes easier to donate an electron as the attraction is weaker
- **Halide ions** are good reducing agents as they can give away the electron that causes the 1- charge
- The reducing ability of the halide ions increases down the group

3b

b) Give the oxidation number for Cl in  $\text{HClO}$  and  $\text{Cl}^-$

- Cl in  $\text{HClO}$  = +1; [1 mark]
- Cl in  $\text{Cl}^-$  = -1; [1 mark]

[Total: 2 marks]

- The oxidation number of the  $\text{Cl}^-$  ion must be -1 as it is an ion by itself
- When working out the oxidation number for Cl in  $\text{HClO}$ 
  - H = +1
  - O = -2
  - Cl must be +1 for the overall charge to be 0

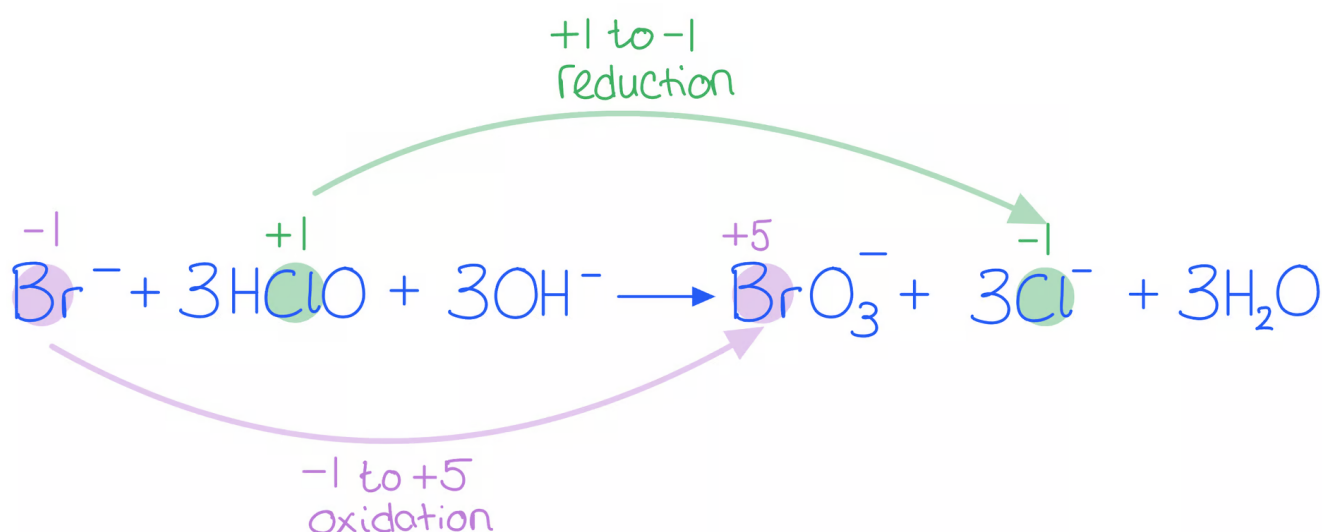
3c

c) The oxidising agent in this reaction is:

- $\text{HClO}$ ; [1 mark]
- Oxidises bromide /  $\text{Br}^-$  from  $-1$  to  $+5$ ; [1 mark]

[Total: 2 marks]

- Oxidation is the loss of electrons, reduction is the gain of electrons
- Writing all of the oxidation numbers on the equation will allow you to identify any changes



- An increase in oxidation number indicates oxidation
- A decrease in oxidation number indicates reduction
- An oxidising agent is itself reduced
- A reducing agent is itself oxidised
  - Therefore the oxidising agent is  $\text{HClO}$
  - Therefore the reducing agent is  $\text{Br}^-$

4

a) The boiling point increases because:

- The boiling point increases due to the increase in  $M_r$  / number of electrons from

$\text{CH}_3\text{CH}_2\text{CH}_2\text{F}$  to  $\text{CH}_3\text{CH}_2\text{CH}_2\text{I}$ ; [1 mark]

- This creates stronger van der Waals forces / London forces / dispersion forces / temporarily induced dipole dipole attractions; [1 mark]

**[Total: 2 marks]**

- Halogenoalkanes will tend to have a higher boiling point from the alkanes from which they are derived
- This is because:
- The  $M_r$  of the molecule increases, therefore van der Waals forces increase so more energy is required to separate the molecules from one another
- There is also increased intermolecular forces as the C-X bonds will have a polar nature due to the difference in electronegativity
- The **main factor governing the trend in boiling point is the  $M_r$  of the molecule:**
- For a given 'R' group the boiling point trend will be
- **$\text{RI} > \text{RBr} > \text{RCI} > \text{RF}$**